Jewoo Suh

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Research Interests: Knowledge Representation & Reasoning, Ontology, Evolutionary & Genetic Algorithms, Mathematical & Computational Sciences, Theoretical Computer Science

Education

University of Oxford, *Christ Church College* 2021.10 – 2025.07
Integrated Master of Mathematics and Computer Science (MMathCompSci)
Thesis: Designing the VG8: An FPGA-Based 8-bit Game Console for Comparative Analysis of 1980s Personal Computers and Game Consoles
Classification: Second Class Honours, Division I (2:1) in Parts A & B; Merit in Part C

Work Experience

LIG Defense & Aerospace (LIG D&A) *Pangyo, Republic of Korea* 2025.07 – Present
Research Engineer @ Technology Innovation Division, AI R&D Team
Serving as Technical Research Personnel (mandatory military service) from 2025.07.

- Develop and maintain the Next Software Framework (NSFW), a multi-OS internal framework deployed across various defence weapons systems for core weapons functionality.
- Collaborate with Palantir Technologies to develop an AI Command & Control solution for Unmanned Surface Vessels and a Multi-Domain Unmanned System Swarm Networking platform.
- Leading development of a company-wide internal AI Toolkit to standardise AI integration across teams.

Xsolla Korea *Seolleung, Seoul, Republic of Korea* 2023.07 – 2024.12
Intern → Process Automation Manager

- Designed and deployed core AWS infrastructure for Xsolla's APAC branch.
- Automated and maintained internal Salesforce database organisation, reducing manual overhead across the regional office.
- Represented Xsolla at Tokyo Game Show 2023 and Busan Indie Connect Festival 2023.

Publications

[1] J. Suh et al., "Iterated Self-Evaluation Impossibility for Large Language Models," *in preparation*, IEEE Access, 2026.

[2] J. Suh et al., "Truthful Task and Resource Allocation in Leader–Follower Agent Organisations," *in preparation*, PRIMA 2026, 2026.

[3] W.-S. Kim et al. (incl. J. Suh), "[Digital Twin-Verified Design and Development of a USV Swarm Command and Control System]," *in preparation*, JKIICE, 2026.

[4] J. Suh, "`.sillok`: A Korean-Specific Lossless Text Encoding Format for Linguistically-Informed Compression and Long-Term Archival," Preprint, 2026.

[5] J. Suh, "Designing the VG8: An FPGA-Based 8-bit Game Console for Comparative Analysis of 1980s Personal Computers and Game Consoles," MMathCompSci Thesis, University of Oxford, 2025.

Projects

Cross-Domain: Cross-field knowledge formalisation via formal publication language. *ongoing*

Language Evolution Simulator: Emergent grammar and lexicon via genetic algorithms. *ongoing*

Voynich Decode Project: Statistical structure analysis of the Voynich Manuscript. *closed*

For further details on publications and projects, visit jewoo-suh.github.io.

Honours & Awards

Minister's Award, Ministry of Science and ICT 2025
2025 Republic of Korea SW Technology Grand Prize, IT Architecture Category

1st Place, Christ Church College Computer Science Presentation Talks 2023

1st Place, Oxford Hack (Oxford University Student Hackathon) 2022

Skills

AI Tools: Claude Code, Gemini, GPT Codex
Programming: Python, Verilog, C, Scala, Unity/C#, Java, Haskell, MATLAB, LaTeX, HLSL
Languages: English (Fluent), Korean (Fluent), Chinese (Conversational), Japanese (Conversational)